Mobile Product Analytics Portfolio • Automated Daily Briefing • *2026-08-22*

Portfolio Executive Summary & Health Dashboard

Multi-App Portfolio Overview (Data Snapshot: August 21, 2026)

Application Name	Bundle ID	Active Users (DAU)	New Users	Total Sessions	Avg Session Duration	Engagement Rate	Crash-Free User Rate	Top Market	Primary Health Status
Music Separation iOS	`com.neuralsound.mss`	143	107	169	333.1s	84.6%	100.0%	South Korea (76.9%)	**■ HEALTHY / SCALING**
Vocal Cleaner	`com.neuralSound.audiocleaner`	3	0	17	3531.1s	88.2%	66.7%	Bangladesh (100.0%)	**■ CRITICAL / UNSTABLE**
Portfolio Aggregate	*N/A*	**146**	**107**	**186**	**625.3s**	**85.0%**	**96.6%**	**South Korea**	**■ DISPARATE PERF.**

Key Cross-App Trends & Portfolio Insights

1. **Massive Growth Discrepancy**: **Music Separation iOS** is experiencing strong organic trajectory in South Korea (+376% DAU growth over 7 days, scaling to 143 DAU), driven by successful top-of-funnel onboarding. Conversely, **Vocal Cleaner** is virtually stagnant (3 DAU, 0 new users), operating essentially as a QA/testing state or abandoned product.
2. **Critical Telemetry & SDK Implementation Defects**:
3. **Slider Debouncing Defect (Music Separation iOS)**: Touch sliders for pitch/tempo controls emit continuous events on every pixel movement, flooding analytics with **570.6 events/user** for pitch changes.
4. **StoreKit Transaction Loop Defect (Vocal Cleaner)**: Duplicate purchase/subscription telemetry indicates a failure to call `Transaction.finish()` or `finishTransaction(_:)`, causing repeated, looping execution of purchase events (12 purchase starts and 6 purchase completions for a single user).
5. **Staged Rollout Alert (Music Separation iOS v1.3.08)**: Build `1.3.08` exhibits a severe drop in engagement rate (**53.8%** vs. **86.8%** baseline in `1.3.07`). Phased release must be halted immediately.

```
PORTFOLIO TRAJECTORY (PAST 7 DAYS) 160 ■ ■■ [MS iOS: 143] 140 ■ ■■■■■■■ 120 ■ ■■■■■■■ 100 ■ ■■■■■■■ 80 ■
■■■■■■■ 60 ■ ■■■■■■■ 40 ■ ■■■■■■■■ ■■■■■■■■ 20 ■ ■■■■■■■■■■■■■■■■■■■■■■■■■ 0
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ [VC: 3] Aug 15 Aug 16
Aug 17 Aug 18 Aug 19 Aug 20 Aug 21
```

Deep-Dive: Music Separation iOS

App Header & Health Score

- **Bundle ID**: `com.neuralsound.mss` | **GA4 Property**: `533563138` | **Stream ID**: `14402502632`
- **App Health Score**: **8.2 / 10**
- *Rationale*: Exceptional onboarding completion rate (95.5%) and strong active user expansion in South Korea. Zero stability crashes reported. Score penalized due to telemetry flood bugs (un-debounced audio player controls) and an engagement regression in the latest staged rollout build (`v1.3.08`).

Old vs. New User Cohort Deep-Dive

Cohort Type	Active Users	Total Sessions	Avg Session Duration	Engagement Rate	Screen / Page Views	Views / Session
New Users	114	112	347.8s	94.6%	229	2.04
Returning Users	41	52	332.9s	69.2%	39	0.75
Unset / System	0	5	6.5s	20.0%	1	0.20

Cohort Analysis: New users demonstrate higher engagement rates (94.6%) and session durations (347.8s) than returning users (69.2%, 332.9s). This indicates high activation efficiency during first-time setup, but highlights potential feature-exhaustion or friction for returning power users who bypass screen views directly to background audio processing.

7-Day Velocity & Trajectory

Date	Active Users	New Users	Sessions	Engagement Rate	Growth Trajectory (DAU)
2026-08-15	30	4	46	82.6%	Baseline
2026-08-16	28	7	38	84.2%	■ -6.7%
2026-08-17	36	14	45	80.0%	■ +28.6%
2026-08-18	57	36	71	83.1%	■ +58.3%
2026-08-19	94	78	120	90.8%	■ +64.9%
2026-08-20	93	65	113	82.3%	■ -1.1%
2026-08-21	143	107	169	84.6%	■ +53.8% (7-Day Total: +376.7%)

User Funnel Diagrams

1. Onboarding & Account Registration Funnel

```
[109] first_open ■ ▼ (100.0%) [111] onboarding_viewed / onboarding_step_viewed ■ ▼ (99.1%) [110]
onboarding_next_tapped ■ ▼ (100.0%) [117] login_screen_viewed ■ ▼ (94.0%) [110] login_succeeded ■ ▼ (96.4%)
[106] first_journey_login_completed ■■■ [OVERALL ONBOARDING CONVERSION: 95.5%]
```

2. Core Feature Execution Funnel (Import to Separation Playback)

```
[105] plus_button_tap / first_journey_add_separation ■ ▼ (50.5% - Major Drop-off) [53] import_started (Files: 78
users | Photos: 63 users) ■ ▼ (88.7%) [47] separation_play_started ■ ▼ (100.0%) [72] separation_player_opened
(includes re-entry)
```

Crash & Stability Health

- **Overall Crash-Free User Rate**: **100.0%**
- **Total Fatal Exceptions**: 0
- **Non-Fatal System Errors**: 0

■ New Build / Staged Rollout Alert Section

```
STAGED ROLLOUT ENGAGEMENT COMPARISON (v1.3.07 vs. v1.3.08) 100% ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
High Baseline (86.8%) 80% ■ 60% ■ ■■■■■■■■■■ Severe Drop (53.8%) 40% ■ ■■■■■■■■■■ 20% ■ ■■■■■■■■■■ 0%
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ Production Baseline (v1.3.07) Staged Rollout
(v1.3.08)
```

Build Version	Active Users	Total Sessions	Engagement Rate	Crash-Free Rate	Staged Rollout Status / Assessment
1.3.07 *(Baseline)*	136	152	**86.8%**	100.0%	■ Production Baseline (93.1% adoption)
1.3.08 *(Staged)*	5	13	**53.8%**	100.0%	■ **ALERT**: -33.0% Engagement drop. Pause rollout!
1.3.06 *(Legacy)*	2	4	**100.0%**	100.0%	■ Deprecated Build

Architectural Recommendation: **PAUSE** the rollout of `1.3.08` immediately. While no crashes were reported, engagement rate collapsed from 86.8% to 53.8%. Investigate potential UI blocking, stem processing hangs, or silent network failures in `1.3.08`.

Top In-App Events & Telemetry Bug Analysis

```
TELEMETRY FLOODING (EVENTS PER USER) player_pitch_changed
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ 570.6 player_tempo_changed
■■■■■■■■■■■■■■■■■■■■■■■■■■ 227.3 player_stem_bal_changed ■■■■■■■■■■■■■■■ 132.0 player_stem_muted ■■■■■
41.8 audio_sessions ■■ 17.3
```

Critical Telemetry Defects Discovered:

1. **Slider Debounce Missing (`player_pitch_changed` & `player_tempo_changed`)**:
2. `player_pitch_changed`: 5,706 events emitted across only 10 users (**570.6 events/user**).
3. `player_tempo_changed`: 2,046 events emitted across only 9 users (**227.3 events/user**).
4. *Root Cause*: Touch interaction events firing continuously during slider drag gestures without continuous interaction debouncing or throttle scheduling.
5. **Stem Balance Event Explosion (`player_stem_balance_changed`)**:
6. 132 events recorded for **1 single user**.

Geographic Distribution

Country / Territory	Active Users	Total Sessions	% of Total Active Users
South Korea	110	126	**76.9%**
Iran	7	9	4.9%
Bangladesh	5	13	3.5%
India	4	2	2.8%
Japan	3	3	2.1%
United States	3	2	2.1%
Other (4 Countries)	11	14	7.7%

Strategic Action Matrix: Music Separation iOS

Priority	Category	Action Item	Root Cause / Target Metric
P0	**Engineering**	**Halt Staged Rollout of v1.3.08**: Pause release in App Store Connect. Investigate 33.0% drop in engagement rate.	Engagement Rate recovery to >85%
P0	**Engineering**	**Fix Analytics Debouncing**: Implement Rx/Combine `.debounce(for: .milliseconds(500))` on pitch/tempo slider control event emitters.	Reduce telemetry volume by ~95%
P1	**Product**	**Optimize Import Conversion**: 49.5% of users tap `plus_button_tap` but fail to trigger `import_started`. Simplify file picker modal UI.	Increase Import Conversion to >75%
P1	**Growth**	**Capitalize on South Korea Surge**: Localize App Store metadata, keywords, and onboarding flow into Korean (`ko_KR`).	Accelerate SK organic growth
P2	**Product**	**Prompt Review Optimization**: 120 `rate_prompt_shown` events hit 119 users early in execution. Delay trigger until *after* second success play.	Higher rating acceptance rate

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Deep-Dive: Vocal Cleaner

App Header & Health Score

- **Bundle ID**: `com.neuralSound.audiocleaner` | **GA4 Property**: `533563138` | **Stream ID**: `15407667671`
- **App Health Score**: **3.5 / 10**
- *Rationale*: Severe acquisition failure (0 new users yesterday), degraded stability (66.7% crash-free rate), and a critical StoreKit analytics retry loop generating duplicated purchase events. Average session duration (~3531s) indicates unhandled background audio execution or stuck processing UI.

Old vs. New User Cohort Deep-Dive

Cohort Type	Active Users	Total Sessions	Avg Session Duration	Engagement Rate	Screen / Page Views	Views / Session
New Users	0	0	0.0s	0.0%	0	0.00
Returning Users	3	17	3531.1s	88.2%	101	5.94

Cohort Analysis: Absolute absence of new user acquisition. Existing returning users display abnormally long session durations (58.8 minutes per session on average), pointing to a runaway background recording loop or an unclosed session state bug.

7-Day Velocity & Trajectory

Date	Active Users	New Users	Sessions	Engagement Rate	Growth Trajectory (DAU)

2026-08-15	3	0	11	63.6%	Flat Baseline
2026-08-16	2	0	9	66.7%	■ -33.3%
2026-08-17	1	0	3	100.0%	■ -50.0%
2026-08-18	2	1	7	85.7%	■ +100.0%
2026-08-19	8	6	17	58.8%	■ +300.0% (Brief Spike)
2026-08-20	1	0	3	66.7%	■ -87.5%
2026-08-21	3	0	17	88.2%	■ +200.0%

User Funnel Diagrams

1. Audio Recording & Denoise Execution Funnel

```
[2] vc_record_open ■ ▼ (100.0%) [2] vc_record_permission_result (Granted) ■ ▼ (100.0%) [2] vc_record_start ■■■
[9 Total Start Events] ■ ▼ (100.0%) [2] vc_record_pause / done ■ ▼ (100.0%) [2] vc_record_denoise_tap ■ ▼ (50.0%
Blocked by Paywall) [1] vc_record_denoise_blocked
```

2. Paywall & StoreKit Transaction Loop (Single User Telemetry)

```
[1 User] paywall_viewed (16 times) ■ ■■■ [1 User] subscribe_tapped (12 times) ■ ■ ■ ■■■ purchase_started (12
times) ■ ■■■ purchase_approved (10 times) ■ ■■■ purchase / purchase_succeeded (6 times) ■■■ ■■ DUPLICATE
PURCHASES ■ ■■■ purchase_failed (5 times) ■ ■■■ purchase_storekit_failed (2 times) ■ ■■■
app_store_subscription_renew (35 events across 2 users)
```

Crash & Stability Health

- **Overall Crash-Free User Rate**: **66.67%** (Critical risk)
- **Total Fatal Exceptions**: **1 user impacted** (`app_exception` logged during recording/denoise pipeline)
- **Impact**: 1 out of 3 daily active users suffered a crash, directly destroying retentive health.

■ New Build / Staged Rollout Alert Section

Build Version	Active Users	Total Sessions	Engagement Rate	Crash-Free Rate	Staged Rollout Status / Assessment
1.0.01	3	17	88.2%	**66.67%**	**■ UNSTABLE PRODUCTION BASELINE**

Alert Statement: No secondary builds are currently in staged rollout. The current production release (`1.0.01`) is deeply unstable (33.3% user crash rate) and contains severe StoreKit integration bugs. An emergency hotfix build (`1.0.02`) is required immediately.

Top In-App Events & Telemetry Bug Analysis

```
STOREKIT RETRY LOOP (EVENTS PER USER) user_engagement ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
55.7 screen_view ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ 50.5 app_store_sub_renew ■■■■■■■■■■■■■■■
17.5 paywall_viewed ■■■■■■■■■■■■■■ 16.0 subscribe_tapped ■■■■■■■■■■■■ 12.0
```

Critical Telemetry & Architectural Defects:

1. **StoreKit Transaction Finish Failure**:

2. A single user generated 12 `purchase_started`, 10 `purchase_approved`, and 6 `purchase_succeeded` events, along with 5 `purchase_failed` events.
3. *Root Cause*: The app fails to call `SKPaymentQueue.default().finishTransaction(transaction)` or StoreKit 2 `transaction.finish()`. Unfinished transactions remain on the queue, causing StoreKit to re-deliver the transaction on every screen transition or app restart, repeatedly firing GA4 purchase events.
4. **Infinite Session Duration Anomaly**:
5. Average session duration is **3,531.1s (~58 minutes)** with 17 sessions spread across 3 users.
6. *Root Cause*: An unclosed audio recording session keeps the iOS background process alive without ending the GA4 session timer properly.

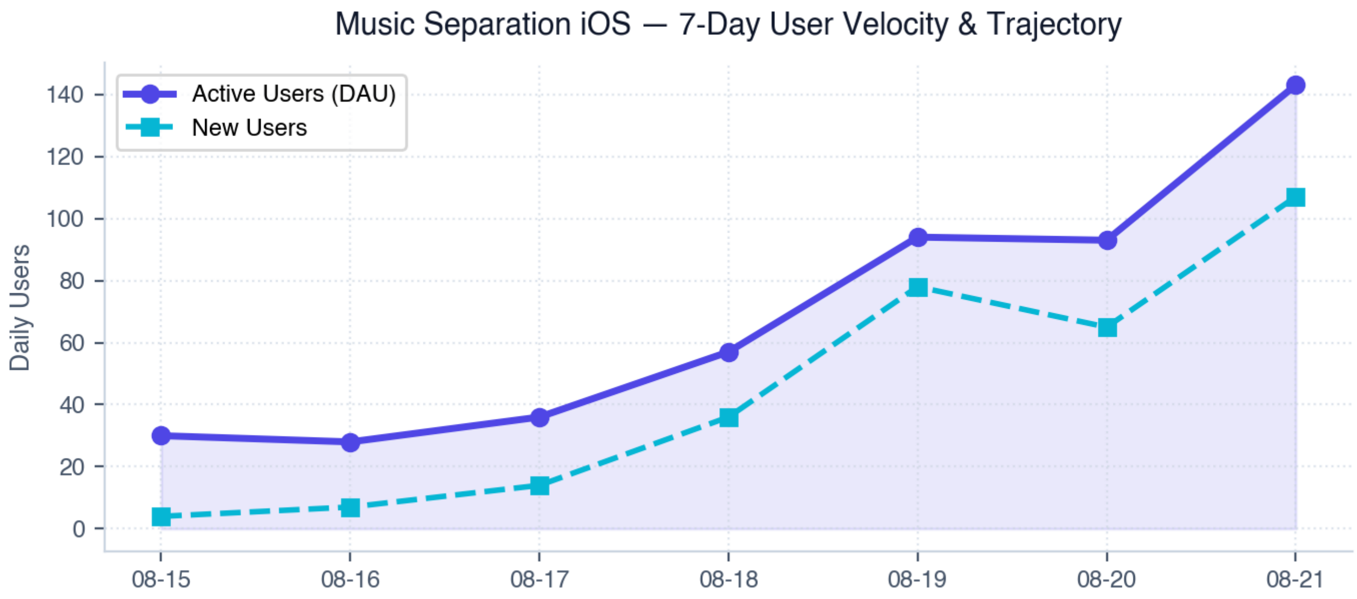
Geographic Distribution

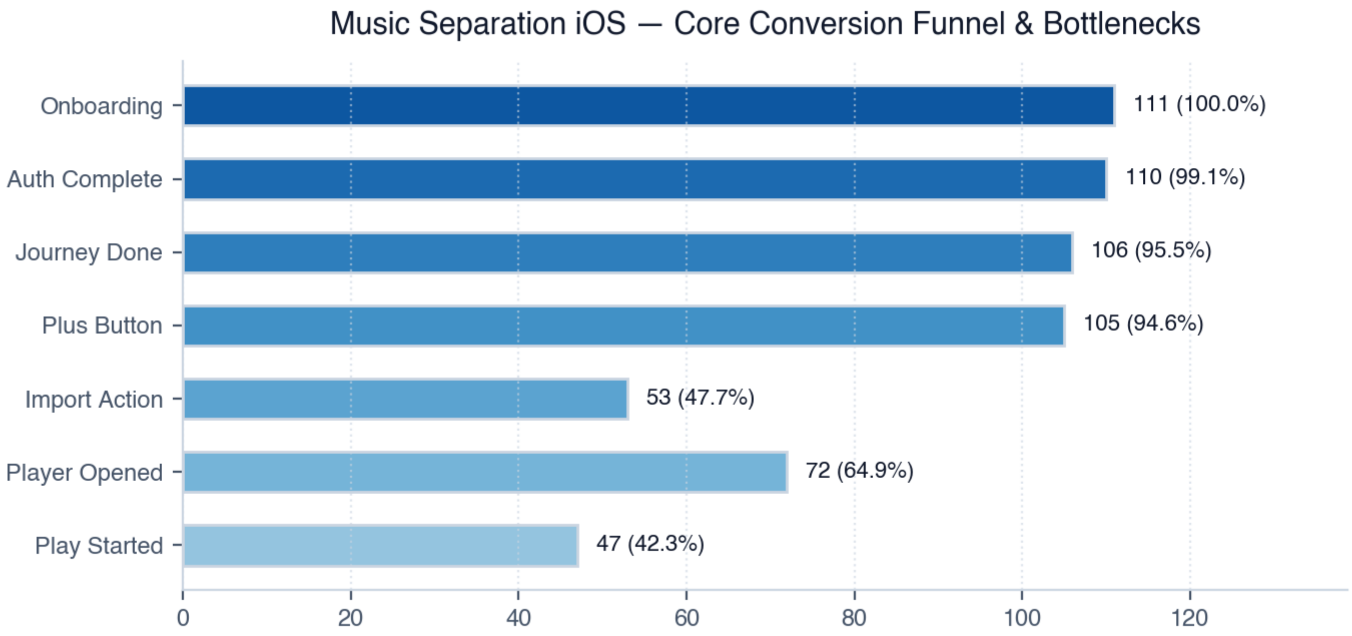
Country / Territory	Active Users	Total Sessions	% of Total Active Users
Bangladesh	3	17	**100.0%**

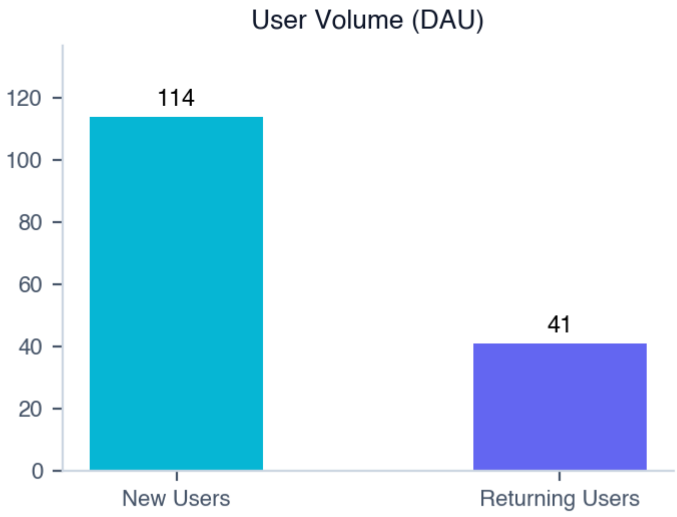
Strategic Action Matrix: Vocal Cleaner

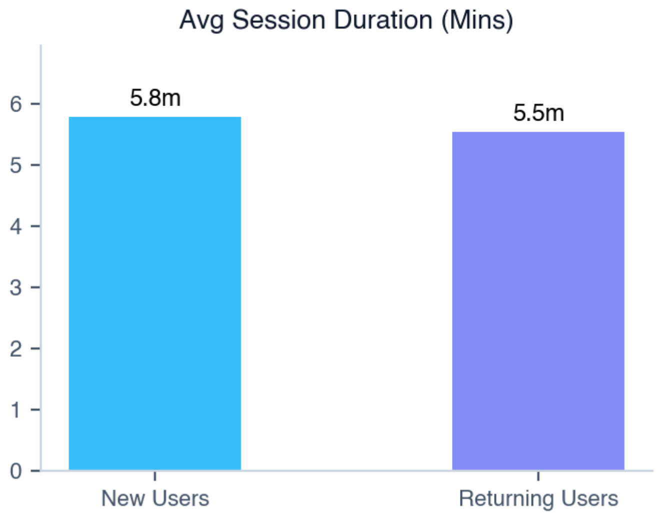
Priority	Category	Action Item	Root Cause / Target Metric
P0	**Engineering**	**Fix StoreKit Transaction Finishing**: Call `transaction.finish()` immediately upon successful processing. Resolve duplicate purchase/renew telemetry.	Eliminate duplicate purchase fires
P0	**Engineering**	**Fix `app_exception` Crash**: Resolve core memory/audio buffer crash in denoise engine affecting 33.3% of users.	Crash-Free Users > 99.0%
P1	**Product**	**Fix Paywall Lockout**: 1 user was hit with 16 paywall views and 2 `vc_record_denoise_blocked` events. Fix subscription status state caching.	Restore seamless premium UI state
P1	**Growth**	**Initiate Basic User Acquisition**: App store optimization (ASO) and localized keyword strategy needed; zero new users acquired naturally.	Re-establish organic installation funnel
P2	**Engineering**	**Background Timer Lifecycles**: Ensure `AVAudioRecorder` background tasks correctly send session pause/stop events to GA4 upon completion.	Normalize average session length metric

■ Visual Graphs & Performance Charts

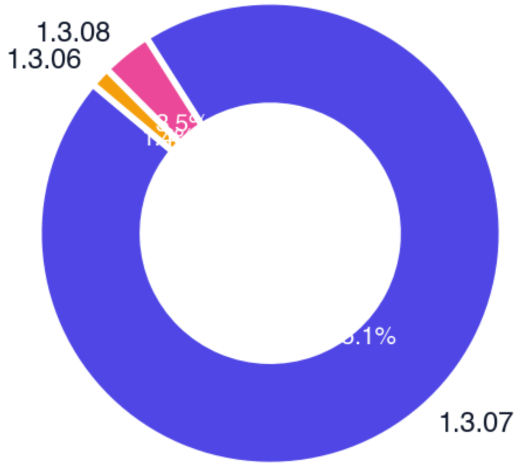

Music Separation iOS — 7-Day User Velocity & Trajectory


Music Separation iOS — Core Conversion Funnel & Bottlenecks

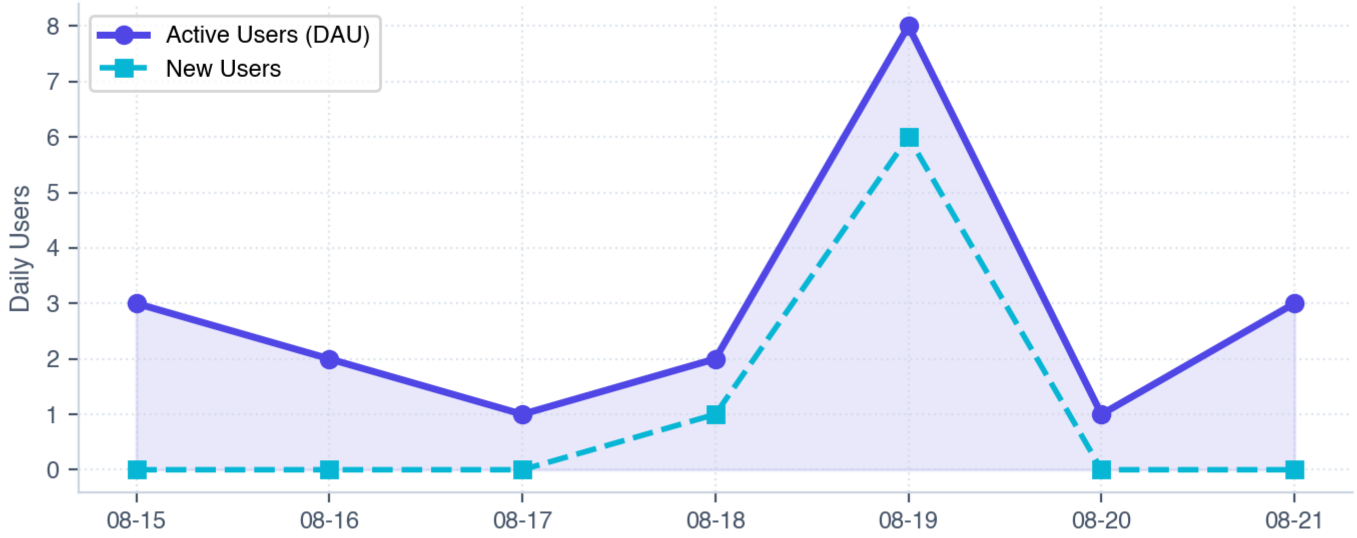

User Volume (DAU)


Avg Session Duration (Mins)

Music Separation iOS — App Version Distribution



Vocal Cleaner — 7-Day User Velocity & Trajectory



Vocal Cleaner — App Version Distribution